

Which $SU(3)$? A didactic tour of Kaluza–Klein model building on the Aloff–Wallach space $X_{1,1}$

Charges, slots, and obstructions — recovering the 1980s technology

kkFable working notes*

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Abstract

In the early 1980s it was briefly reasonable to hope that the Standard Model was pure geometry: gauge bosons from isometries of a small seven-dimensional space, charges from its bundles, couplings from its shape. The hope died for specific, instructive reasons that are no longer taught. We revisit that technology on one concrete carrier — the Aloff–Wallach space $X_{1,1} = SU(3)/U(1)_{1,1}$, an $SO(3)$ bundle over $\mathbb{CP}^2$ — and work one sharp question end to end: *which $SU(3)$ does the base carry, the electroweak one (Option A) or QCD colour (Option B)?* The answer is a tour of everything a young model builder needs from the old toolbox: the isometry→gauge-boson rule, Weinberg’s coupling-from-inertia formula, coset harmonics and equivariant bundles, spin^c structures and charge parity, index-theorem generation counting — each with the explicit Lagrangian it descends from. Option B turns out to reproduce the Standard Model’s charge lattice with surgical near-misses: five of seven matter slots land exactly, the electron singlet hides in the gravitino sector at the canonical-bundle twist, the Higgs is forced out of scalar harmonics into internal tensor components, and two obstructions remain irreducible — there is no room for a dynamical hypercharge boson, and no chirality. Chasing those two through the dimensional ladder $D5-D6-D7$ (total spacetime 9/10/11) reveals an electroweak alternation, a charge-to-flux conversion law, and a transition whose endpoint realizes, at once, the two ways a gauge sector can leave a theory: the vacuum running to infinity, or the coupling. We keep $c = 1$ but write $\hbar$ and $\kappa^2 = 8\pi G$ explicitly throughout, so the reader can see which statements are classical geometry and which are quantum spectroscopy.

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*Compiled from the derivation notes N0–N12 of the kkFable workspace (2026-06-11/12); every load-bearing claim is re-derived by hand there, with two independent verification passes recorded. This document is pedagogical: it narrates, the notes prove.

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1 Introduction: a lost subject

Between roughly 1979 and 1985 a substantial part of the high-energy community tried to derive the Standard Model from pure gravity in eleven dimensions. The programme had crisp early victories. Witten observed that seven is the *minimal* number of extra dimensions whose isometries can contain $SU(3) \times SU(2) \times U(1)$, and that eleven is also the maximal dimension for supergravity — a coincidence that still deserves a moment of silence [1]. The seven-manifolds that realize the full Standard Model gauge group were classified (the M^{pqr} spaces, circle bundles over $\mathbb{CP}^2 \times S^2$ [1, 3, 7]); Weinberg showed that gauge couplings are computable moments of inertia of the internal shape [2]; and the machinery of coset harmonics turned spectroscopy into representation theory [4, 5].

Then came the three deaths, all instructive: no chirality (massless fermions on smooth odd-dimensional internal spaces pair left with right [1, 6]); no usable vacuum (the natural ground states are anti-de Sitter at the compactification scale); and no Standard-Model matter in the harmonic towers (the M^{pqr} spectra simply do not contain the quark and lepton representations [7]). The community moved on to strings, and the specific *techniques* — how to read charges off a coset, where hypercharge actually lives, why the Higgs refuses to be a scalar harmonic — quietly left the curriculum.

This paper is a guided tour of those techniques on a single, particularly instructive carrier, with a single organizing question. The carrier is the Aloff–Wallach space

$$X_{1,1} = SU(3)/U(1)_{1,1}, \quad U(1)_{1,1} = \{ \text{diag}(z, z, z^{-2}) \}, \quad (1)$$

a seven-dimensional homogeneous space fibred as $SO(3) \rightarrow X_{1,1} \rightarrow \mathbb{CP}^2$. It is, in a precise sense, the *other* natural seven-manifold built from $SU(3)$: where the M^{pqr} spaces keep a hypercharge circle and pay for it elsewhere, $X_{1,1}$ trades that circle away for a fibre with the shape of the weak group. The question is the fork that names this paper:

*The base $\mathbb{CP}^2 = SU(3)/U(2)$ has isometry $SU(3)$ and isotropy $U(2)$.
 Is that $SU(3)$ the electroweak $SU(3)$ of a 3-3-1 model (Option A),
 or is it QCD colour, with the fibre $SO(3)$ as the weak group (Option B)?*

Option A was the assumption of all earlier work on this geometry; Option B is the flip. Resolving the flip turns out to exercise every tool in the 1980s box exactly once, and the near-misses are more instructive than the failures. Everything below is elementary — the point is precisely that it is elementary and yet nobody under fifty seems to know it.

Units. We set $c = 1$ and keep $\hbar$ and the gravitational coupling $\kappa^2 = 8\pi G$ *visible*. The payoff is didactic: formulas containing only κ are classical geometry (couplings, masses of gauge bosons from shapes); $\hbar$ enters exactly where momentum is quantized on a compact direction (Kaluza–Klein masses, charge quantization) and in loop quantities like $\alpha_a = \hbar g_a^2/4\pi$. Watching where $\hbar$ does *not* appear is half the lesson.

2 The toolbox: Kaluza–Klein in five moves

2.1 Move 1: isometries become gauge bosons

Take $(4 + d)$ -dimensional gravity on $M_4 \times X_d$,

$$S = \frac{1}{2\hat{\kappa}^2} \int d^{4+d}x \sqrt{-\hat{g}} \hat{R}, \quad \hat{\kappa}^2 = 8\pi\hat{G}. \quad (2)$$

If X_d has isometry group G_{iso} with Killing vectors K_a , the metric fluctuation

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu + g_{mn}(y) (dy^m + K_a^m(y) A_\mu^a(x) dx^\mu) (dy^n + K_b^n(y) A_\nu^b(x) dx^\nu) \quad (3)$$

is consistent at the massless level, and inserting it into (2) produces four-dimensional Einstein–Yang–Mills theory: *one massless gauge boson per Killing vector, gauge group = the isometry group*. No further choice is available: you cannot order more gauge symmetry than the shape possesses. This single sentence already decides a large part of our fork (Section 4).

Example 1: the original circle, with $\hbar$ and κ in their places

$X = S^1$ of circumference L , coordinate $y \in [0, L)$, one Killing vector ∂_y :

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu + \rho^2 (dy + A_\mu dx^\mu)^2.$$

Insert into (2) with $d = 1$ and integrate over y :

$$S_4 = \int d^4x \sqrt{-g} \left[\frac{\rho L}{2\hat{\kappa}^2} R - \frac{\rho^3 L}{8\hat{\kappa}^2} F_{\mu\nu} F^{\mu\nu} + \frac{L}{2\hat{\kappa}^2} (\text{gradient terms for } \rho) \right].$$

Reading off: $\kappa^2 \equiv \hat{\kappa}^2/(\rho L)$ and $\frac{1}{g^2} = \frac{\rho^3 L}{2\hat{\kappa}^2}$ — *both classical: no $\hbar$ anywhere*. Quantum mechanics enters through the modes $\phi_n \propto e^{2\pi i n y/L}$: their 4D masses and charges are

$$m_n = \frac{2\pi\hbar|n|}{\rho L}, \quad q_n = n \quad (\text{in units of } g),$$

i.e. $\hbar$ appears exactly once, multiplying the quantized circle momentum. Charge quantization is momentum quantization — Klein's 1926 observation, still the cleanest sentence in the subject.

2.2 Move 2: couplings are moments of inertia (Weinberg's rule)

For a general X_d the same reduction gives, per isometry direction a ,

$$\boxed{\frac{1}{g_a^2} = \frac{1}{2\hat{\kappa}^2} \int_{X_d} \sqrt{g_X} g_{mn} K_a^m K_a^n \equiv \frac{I_a}{2\hat{\kappa}^2}} \quad (4)$$

— the *moment of inertia* of the internal shape about the axis a [2]. Two consequences organize everything later:

- If two generators sit inside one *simple* isometry factor, Schur's lemma forces their inertias into a fixed ratio: the corresponding coupling ratio is *group theory*, not geometry. (This is how Option A predicts $\sin^2 \theta_W = 1/4$.)
- If they sit in *different* factors of a semisimple isometry group, the ratio is a free shape modulus. (This is how Option B loses that prediction.)

2.3 Move 3: spectroscopy is representation theory (Frobenius)

The one formula to remember

Fields on $X = G/H$ valued in the homogeneous bundle of the H -module W , twisted by a background flux line $\mathcal{O}(m)$, have 4D mode content

$$\text{modes} = \bigoplus_{\lambda} V_{\lambda} \otimes \text{Hom}_H(V_{\lambda}, W \otimes \mathcal{O}(m)),$$

i.e. a mode in the G -irrep V_{λ} exists iff V_{λ} , restricted to H , contains the H -type of the bundle. *All* charge bookkeeping below is this one line, evaluated on small tables. For $X_{1,1}$, $H = U(1)_{1,1}$ is tiny, so the condition is a single charge balance: $q_8(\text{state}) + q_8(W\text{-component}) + m = 0$, where q_8 is the eigenvalue of $\text{diag}(1, 1, -2)$.

2.4 Move 4: fermions see double covers (and a one-line parity law)

Spinor fields transform under the *double cover* of the isometry/isotropy groups: they may carry half-integer labels that bosons cannot. On $X_{1,1}$ this is governed by one group element, which deserves a box because it will do a remarkable amount of work:

The parity element ζ

Let $\zeta = \exp(i\pi \text{diag}(1, 1, -2)) = \text{diag}(-1, -1, 1) \in SU(3)$. This single element lies *both* in the quotient circle $U(1)_{1,1}$ (at $z = -1$) *and* in the upper-block $SU(2)$ (as its centre $-\mathbf{1}_2$). Evaluate it twice on any weight state of any $SU(3)$ representation:

$$(-1)^{q_8} = \zeta = (-1)^{2I} \implies \boxed{q_8 \equiv 2I \pmod{2}}.$$

Doublet-ness and charge parity are locked: every $SU(2)$ doublet carries odd q_8 , every singlet/triplet even q_8 . Geometrically this $\mathbb{Z}_2$ is simultaneously (i) the obstruction to lifting the fibre $SO(3)$ to $SU(2)$, (ii) the non-spin-ness of $\mathbb{CP}^2$ repaired by odd flux (the spin^c condition), and (iii) — as we will see — the seed of the Standard Model's $\mathbb{Z}_6$ charge correlation.

2.5 Move 5: the chirality wall

On a smooth, compact, *odd*-dimensional internal space the internal Dirac operator has no chirality grading; its modes pair into 4D left–right couples with identical quantum numbers. The spectrum is vector-like, no matter how clever the bundles [1, 6]. Twisting by spin^c lines does not evade it. Every slot statement in this paper is therefore a statement about *vector-like pairs*; producing a chiral spectrum needs an even-dimensional stage (Section 6) or singular geometry [14]. Keeping this wall visible at all times was a 1980s discipline worth reviving.

3 The carrier: $X_{1,1}$ in one section

Three descriptions of one space, each carrying one physics lesson:

(a) As a circle quotient. $X_{1,1} = SU(3)/U(1)_{1,1}$. The quotient circle is generated by $H_8 = \text{diag}(1, 1, -2)$ — the *hypercharge-shaped* direction of $SU(3)$ (it is the centre of the block

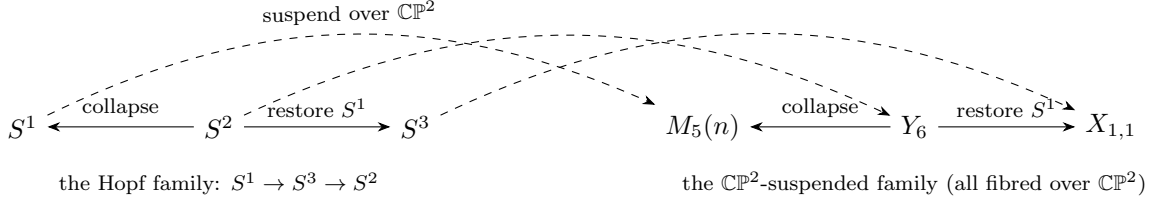


Figure 1: The dimensional ladder. Left: the familiar Hopf triple. Right: its suspension over $\mathbb{CP}^2$: the lens-space circle bundle $M_5(n) = S(\mathcal{O}(n)) \simeq S^5/\mathbb{Z}_n$ (internal $D5$, total spacetime 9), the flag manifold $Y_6 = SU(3)/T^2$ (an S^2 bundle over $\mathbb{CP}^2$; internal $D6$, total 10), and the Aloff–Wallach space $X_{1,1}$ (internal $D7$, total 11).

subgroup $U(2) = S(U(2) \times U(1))$). Lesson: the circle you divide out acts trivially on the quotient; *whatever physics you wanted from it can only survive as bundle data, never as a gauge boson.*

(b) As a bundle. The right action of $U(2)/U(1)_{1,1} = PU(2) \cong SO(3)$ survives the quotient and rotates the fibres of $SO(3) \rightarrow X_{1,1} \rightarrow \mathbb{CP}^2$: a fibre shaped like the (adjoint of the) weak group over a base shaped like a colour space. The full isometry group is $SU(3) \times SO(3)$ (Wilking), of rank $2 + 1 = 3$. The tangent space splits into the fibre block m_0 (3 directions, $q_8 = 0$, metric scale x_1) and the base block m_3 (4 directions, $q_8 = \pm 3$, scale x_2); the shape modulus $t = x_1/x_2 - 1$ has two Einstein points, $t = +1$ (3-Sasakian) and $t = -3/5$ (squashed) — remember those two wells, they return in Section 6.

(c) As a rung. Inside the maximal torus $T^2 \subset SU(3)$ live two orthogonal circles: the Y -circle (H_8) and the T_3 -circle ($H_3 = \text{diag}(1, -1, 0)$). The three spaces of Figure 1 are the three ways of distributing them between geometry and flux:

rung	gauged (isometry)	flux (bundle label)	rank
$D5: M_5(n)$	$SU(3) \times U(1)_Y$	weak data absent	3
$D6: Y_6$	$SU(3)$ only	both: base flux n , monopole r	2
$D7: X_{1,1}$	$SU(3) \times SO(3)_{\text{weak}}$	hypercharge flux n	3

On no rung are both electroweak circles geometric — the alternation that drives Section 6. The Standard Model needs rank 4 on one space; this family caps at 3.

4 The fork, part I: gauge bosons and the rank budget

Under Option A the electroweak pair (T_3, Y) sits inside the simple base $SU(3)$: rank 2 of the available 3, with the fibre $SO(3)$ left over as a custodial spectator. The budget closes, at the price of having no colour anywhere.

Under Option B, colour consumes the whole $SU(3)$ (both Cartans are gluons — and note that no hypercharge can hide inside colour: $Z(\mathfrak{su}(3)) = 0$), the weak group consumes the fibre Cartan, and the count reads

$$\underbrace{2}_{\text{gluons}} + \underbrace{1}_{W^3} + \underbrace{1}_B = 4 > 3 = \text{rank}(SU(3) \times SO(3)).$$

One boson is missing, and it is precisely hypercharge. The would-be Y circle is the quotient circle of description (a): trivial action, no Killing vector, no gauge field. The candidate repairs all die

by short arguments: the bare double cover $SO(3) \rightarrow SU(2)$ adds no Cartan; a Betti $U(1)$ from the 3-form on the harmonic 2-form ($b_2(X_{1,1}) = 1$) exists in supergravity embeddings but *every Kaluza–Klein state is neutral under it* — a fact the M^{11} harmonic analysis states verbatim [7].

Example 2: the Lagrangian that is missing a term

Reducing (2) on $(X_{1,1}, g_t)$ with the rule (4) gives, under Option B,

$$\mathcal{L}_4 = \frac{1}{2\kappa^2} R - \frac{x_2 V_7}{8\hat{\kappa}^2} \text{tr} G_{\mu\nu} G^{\mu\nu} - \frac{x_1 V_7}{8\hat{\kappa}^2} \text{tr} W_{\mu\nu} W^{\mu\nu} - \frac{M_P^2}{2} (\partial\sigma)^2 + \dots$$

(G : gluons; W : the fibre $SO(3)$; σ : moduli). The term $-\frac{1}{4}g_Y^{-2}B_{\mu\nu}B^{\mu\nu}$ *does not exist*: there is no eleventh Killing vector to generate it. Every formula of electroweak phenomenology that divides by g' divides by zero on this carrier. Note also what the two visible coefficients say: $g_2^2/g_3^2 = x_2/x_1$ is a free shape ratio (two factors, no Schur lock), so even colour–weak unification statements are moduli, not predictions.

The same calculus kills the angle. Counterfactually granting a Y boson normalized on the base block with strength k_Y ,

$$\sin^2 \theta_W = \frac{g_Y^2}{g_2^2 + g_Y^2} = \frac{x_1}{x_1 + k_Y x_2} = \frac{1+t}{(1+t) + k_Y}, \quad (5)$$

monotone in the free modulus t : every value in $(0, 1)$ is attained. Contrast Option A, where both generators live in one simple factor and Schur fixes $\sin^2 \theta_W = \text{tr} T_3^2 / \text{tr} Q^2 = 1/4$ on the lepton triplet, metric-independently. This is the cleanest classroom example we know of Weinberg’s dichotomy: *same geometry, same formula (4), and the answer flips from “rigid number” to “free dial” purely according to where the generators sit.*

5 The fork, part II: the charge lattice and the matter slots

Now the spectroscopy. Apply the Frobenius box with the branching data of $SU(3)$ ($\mathbf{3} = \mathbf{2}_{+1} \oplus \mathbf{1}_{-2}$ under $SU(2) \times U(1)_8$, etc.), the spinor module

$$\Delta_7 = \mathbf{2}_{+3} \oplus \mathbf{2}_{-3} \oplus \mathbf{3}_0 \oplus \mathbf{1}_0,$$

and the normalization $Y = m/6$ for the flux charge m (forced, not chosen: see below). Under Option B the 4D colour representation is the $SU(3)$ label λ , the weak isospin I_w is the combined $SU(2)$ spin, and one finds (notes N6):

SM slot	(colour, I_w , Y)	twist m	status on $X_{1,1}$ (Dirac sector)
Q_L	$(\mathbf{3}, \frac{1}{2}, +\frac{1}{6})$	∓ 1	exists
u_R	$(\mathbf{3}, 0, +\frac{2}{3})$	∓ 4	exists
d_R	$(\mathbf{3}, 0, -\frac{1}{3})$	± 2	exists
L	$(\mathbf{1}, \frac{1}{2}, -\frac{1}{2})$	± 3	exists (<i>fixes the unit $Y = m/6$</i>)
ν_R	$(\mathbf{1}, 0, 0)$	0	exists
e_R	$(\mathbf{1}, 0, -1)$	± 6	absent (selection rule)
H	$(\mathbf{1}, \frac{1}{2}, +\frac{1}{2})$, scalar	—	absent (scalar fields)

Five slots land on exact Standard-Model hypercharges with a single normalization constant; the geometry even enforces the Standard Model's global $\mathbb{Z}_6$ structure,

$$6Y \equiv 2t(\lambda) + 3(2I_w) \pmod{6} \quad (6)$$

(t = triality) — the ζ -parity law and the triality lock combined, which is precisely the charge correlation that makes the SM gauge group $[SU(3) \times SU(2) \times U(1)]/\mathbb{Z}_6$. And then two slots fail, by clean selection rules: a colour-singlet weak-singlet fermion is pinned to $Y = 0$ (so no e_R), and a colour-singlet weak-doublet *scalar field* does not exist for any twist (so no elementary Higgs). Selective failure of exactly two occupants of an otherwise perfect lattice is the signature of structure, not accident — and indeed both absences have mechanisms, and both mechanisms have escapes.

5.1 The electron hides in the gravitino (charge doubling)

The Dirac-sector bound is $|q_8(\Delta_7)| \leq 3$, and worse, $I_w = 0$ forces the neutral component. But eleven-dimensional supergravity offers one more fermionic bundle: the *internal gravitino components* ψ_m , which are 4D spin- $\frac{1}{2}$ fields valued in $W = \Delta_7 \otimes m$. Decomposing (notes N11, independently verified):

$$\Delta_7 \otimes m \supset \mathbf{1}_{+6} \oplus \mathbf{1}_{-6} \quad (\text{exactly one each}),$$

because the spinor and tangent bundles of $X_{1,1}$ share the same charged doublet $\mathbf{2}_{\pm 3}$, and $\mathbf{2} \otimes \mathbf{2} \rightarrow \mathbf{1}$ adds the charges: $3+3=6$. The resulting slot is $(\mathbf{1}, 0, Y = \mp 1)$ at twist $m = \pm 6$: the right-handed electron, at *twice the lepton-doublet twist* — in the same unit $Y = m/6$ fixed by L , with no new freedom. Three remarks make this more than a curiosity:

1. *It is the canonical-bundle locus.* On $\mathbb{CP}^2$ alone, Dolan and Nash found e_R at the twist $q = -3$, the canonical bundle K [8]; the $X_{1,1}$ slot at $|m| = 6$ is its 7D avatar ($\Lambda^2 \mathbf{2}_{+3} = \mathbf{1}_{+6}$). The electron singlet persistently sits where the spin^c geometry is most twisted.
2. *The mechanism is rigid.* Doubling needs the tangent and spinor bundles to share one odd-charged doublet. On $S^5 \times S^2$ the tangent charge is too small and the doubled charges come out as doublets, not singlets: no e_R there, verifiably. On M^{11} the *only* massless charged colour/weak singlets in the entire computed spectrum are the two gravitini [7]. Across every carrier examined: *if this family has an electron, it is gravitino-borne.*
3. *It survives gauge fixing.* The γ -trace and longitudinal pieces of ψ_m remove one Δ_7 's worth of states, all bounded by $|q_8| \leq 3$: the ± 6 singlets are untouchable.

5.2 The Higgs is forced into the internal geometry

The scalar no-go has the same texture. A 4D scalar descending from a scalar *field* on $X_{1,1}$ sits in $W = \mathbf{1}$: colour-singlet $\Rightarrow$ charge balance at $q_8 = 0 \Rightarrow$ integer I_w . No doublet, ever. But scalars also descend from *internal tensor components* — and there the doublets reappear:

$$\Lambda^3 m \supset 2 \cdot \mathbf{2}_{\pm 3} \oplus \mathbf{4}_{\pm 3}, \quad \text{Sym}^2 m \supset \mathbf{2}_{\pm 3} \oplus \mathbf{4}_{\pm 3},$$

so the 3-form components A_{mnp} and the mixed metric components $g_{(fibre, base)}$, at twist $|m| = 3$, furnish colour-singlet $I_w = \frac{1}{2}$ scalars at $Y = \pm \frac{1}{2}$: Higgs quantum numbers. The lesson generalizes into a slogan the 1980s learned the hard way and we can now state as a selection rule: *the Standard-Model Higgs cannot be a scalar harmonic; if geometry provides it at all, it provides it as an internal*

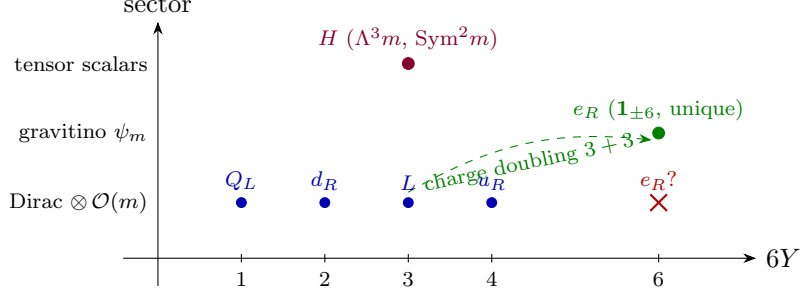


Figure 2: The colour-singlet charge ladder on $X_{1,1}$ under Option B (charges in units $6Y = m$). The Dirac sector reaches $|6Y| \leq 3$ for singlets and places five SM slots; e_R at $|6Y| = 6$ is reachable only by the gravitino components (charge doubling through the shared doublet $\mathbf{2}_{\pm 3}$), and the Higgs only by internal tensor scalars at $|6Y| = 3$.

component of a higher-spin field — which is exactly the ansatz of coset-space dimensional reduction and of modern gauge–Higgs unification. What looked like a modelling preference is a theorem about charge locking.

5.3 Scorecard and the two irreducible obstructions

With the supergravity field menu — and nothing exotic added — *all seven* SM slots exist on $X_{1,1}$ under Option B. What remains impossible is carrier-level, not slot-level:

1. **No dynamical hypercharge boson** (the rank budget of Section 4);
2. **No chirality** (Move 5: every slot above is a vector-like pair).

For comparison, the rank-complete carriers fail in the mirrored way: M^{111} gauges the full $SU(3) \times SU(2) \times U(1)$ but its harmonic lattice contains *no* $SU(2)$ doublets at any level and *no* charged colour-singlet matter [7]; $S^5 \times S^2$ gauges rank 4 but lands only Q_L and L . We call this pattern *complementary exclusion*: within the whole $SU(3)$ -based family, gauge-rank completeness and the doublet/lepton matter mechanism never coexist. The literature’s one working SM spectrum from these spaces — Dolan–Nash’s — evades the dichotomy by gauging *holonomy* instead of isometry and paying with non-dynamical gauge fields [8, 9]; under an isometry reading, their construction is exactly Option-A-shaped. Colour on the $\mathbb{CP}^2$ isometry is, in a precise sense, the one thing this geometry refuses to subsidize fully.

6 The ladder: transitions, the two exits, and criticality

The two irreducible obstructions live on *different rungs* of Figure 1: $D5$ has the hypercharge boson; $D6$ is even-dimensional and can carry a chiral index (this is where the generation count $\nu_n = (n^2 - 1)/8$ lives — Appendix B); $D7$ has the weak group and the matter slots. The Standard Model, on this family, is *phase-distributed*. That makes the transitions between rungs the central object, and here the 1980s technology connects to thoroughly modern questions.

6.1 The $D7 \rightarrow D6$ collapse is a quintet Higgsing

Inside the invariant metric family of $X_{1,1}$, split the fibre block: $x_{\parallel}$ on the T_3 axis, $x_{\perp}$ on the rest. The space of fibre metrics is $\text{Sym}^2(\mathbf{3}) = \mathbf{1} \oplus \mathbf{5}$ of the weak $SO(3)$: the breathing mode and a

quintet. The axis squash is the $I_3 = 0$ component of the quintet, so the collapse family is, in 4D language, a vacuum line of a colour-singlet, $I_w = 2$, $Y = 0$ scalar Φ_5 — an order parameter that breaks $SU(2)_L \rightarrow U(1)_{T_3}$ geometrically:

Example 3: the Lagrangian of a collapsing dimension

Along the family $g(x_{\parallel}, x_{\perp}, x_2)$ the reduction gives (schematically but with every coefficient's origin visible; $V = \sqrt{x_{\parallel} x_{\perp}^2 x_2^4} \cdot v_0$)

$$\mathcal{L}_4 = \frac{1}{2\kappa^2} R - \frac{x_{\parallel} V}{4\hat{\kappa}^2} F_{\mu\nu}^{(T_3)} F^{(T_3)\mu\nu} - \frac{x_{\perp} V}{4\hat{\kappa}^2} |F_{\mu\nu}^{(W^{\pm})}|^2 - \frac{M_P^2}{2} (\partial\sigma)^2 - M_W^2(\sigma) W_{\mu}^{+} W^{-\mu} - V(\sigma) + \dots$$

with $\sigma \propto M_P \log(x_{\perp}/x_{\parallel})$ the canonical quintet modulus and $M_W^2 \propto g_2^2 \langle \Phi_5 \rangle^2$. The T_3 boson stays *exactly massless* at every $x_{\parallel} > 0$ (it is an isometry of the whole family); the $W^{\pm}$ are Higgsed by an $I_w = 2$ vev — which is *not* the Standard-Model mechanism and protects nothing; it is the ladder's own, geometric, Higgsing.

Now send $x_{\parallel} \rightarrow 0$ (Gromov–Hausdorff collapse $X_{1,1} \rightarrow Y_6$, curvature bounded) and watch the gauge sector leave the theory. It leaves *twice, in the two inequivalent ways a gauge sector can leave*:

- **The vev exit.** For $W^{\pm}$: g_W stays finite while $\sigma \rightarrow \infty$ — infinite distance in moduli space, $M_W \rightarrow \infty$. The broken pair decouples because *the vacuum runs to infinity*.
- **The coupling exit.** For the surviving $U(1)_{T_3}$: massless the whole way, but $1/g_{T_3}^2 \propto x_{\parallel} V \rightarrow 0$. The boson leaves at *zero mass through infinite coupling*.

One transition, both exits, assigned by representation: broken non-abelian directions exit through the vacuum, the surviving abelian through the coupling. Meanwhile the *quantum numbers* cross the transition smoothly: collapsing the circle converts the gauged T_3 charge of every mode into the monopole flux label of the limit,

$$r = 2I_3, \tag{7}$$

which is precisely the vertical twist of the $D6$ fermion construction. Charge lattices transport; gauge dynamics does not. The open dynamical question of this whole programme is now locatable in one sentence: *what is the strong-coupling fate of the abelian factor at the endpoint — confinement, duality, or absorption into the flux sector?*

6.2 Criticality: when is a jump a jump?

A Higgs field breaks a symmetry only when its potential *has the Mexican-hat shape*; a parameter sliding continuously along a fixed potential changes nothing qualitative. The same discipline applies to dimensions. Moving along the collapse family at $x_{\parallel} > 0$ is sliding a vev: nothing critical happens. The endpoint is different in kind: the dimension drops, the isometry group jumps, and an *infinite tower* of states ($M_r \sim \hbar|r|/R$, $R \propto \sqrt{x_{\parallel}}$) degenerates at once. That tower is exactly what distinguishes a genuine dimensional transition from a mere symmetry restoration: Witten's IIA/M-theory jump supplied the benchmark (a radius growing with a modulus, an infinite charge family, masses $\sim \hbar|n|/R$) [13], and the restoration $D6 \rightarrow D7$ satisfies all three ingredients *literally*, because the circle and its momentum tower are real geometry here. In total dimension this is a $10 \rightarrow 11$ -type jump: continuous control, critically discontinuous endpoint.

$$\begin{array}{ccccc}
M_5(n) & \xleftarrow[S^2 \rightarrow S^1 \text{ (open half)}]{} & Y_6 & \xrightarrow[\text{restore } S^1: \text{ tower } M_r \sim \hbar|r|/R]{\text{charge} \leftrightarrow \text{flux: } r = 2I_3} & X_{1,1} \\
SU(3) \times U(1)_Y & \xleftarrow[\text{boundary/inflow reading}]{} & SU(3) \text{ only; } (n, r) \text{ flux} & \xrightarrow{} & SU(3) \times SO(3)_W \\
\text{total } D = 9 & & \text{total } D = 10 & & \text{total } D = 11
\end{array}$$

Y boson exists only here;
at $D6/D7$ it is flux n .

chiral index lives here:
 $\nu_n = (n^2 - 1)/8$ generations.

$D7 \rightarrow D6$ exits:
 $W^\pm$: $\sigma \rightarrow \infty$ (vev)
 $U(1)_{T_3}$: $g \rightarrow \infty$ (coupling)

Figure 3: The electroweak alternation and the transitions. Restoration $D6 \rightarrow D7$ passes the M-theory-style tower test by construction (radius modulus, infinite flux family, gap closing as $\hbar|r|/R$); the collapse realizes the two exits; the $D6 \leftrightarrow D5$ half remains open, organized by the same vev-vs-coupling dichotomy.

Where is the hat itself? The potential $V(\sigma)$ on the invariant moduli is not flat: on the t -axis it has *two* known stationary points — the two Einstein metrics ($t = +1$ and $t = -3/5$) — and the collapse direction is an infinite-distance runaway, not a minimum. The sharp open problem, stated so a student could attack it: compute the reduced potential $V(x_\parallel, x_\perp, x_2)$ (its t -axis restriction is already in the prior literature of this programme), and determine which deformations — flux n , curvature couplings, a cosmological term — move, merge, or create its wells. That is the hat-formation analysis for a dimension, with the two Einstein points as its first data.

6.3 Two views of one transition: collapse and emergence

There is a question the reader should now ask, because Witten taught us all to ask it: we *derived* the transition downward ($D7 \rightarrow D6$, a collapse), but the famous version runs upward — IIA string theory discovering its eleventh dimension as the coupling grows [13]. Are both views suitable here? Yes — and the asymmetry between them is the last lesson of the tour.

The collapse view takes the geometry as fundamental: the momentum tower $M_r \sim \hbar|r|/R$ decouples upward, the flux labels are the frozen memory of eaten charges. Every step is a theorem of the metric family.

The emergence view takes the $D6$ description as fundamental: one observes flux sectors labelled by r , notices their spectrum organizing into a tower whose gap closes as a modulus grows, and *infers* a circle. This is exactly Witten's logic — but his version had a hidden ingredient. The IIA D0-brane tower is BPS: $M_n = |n|/(g_s \ell_s)$, exactly linear in the charge, protected all the way to strong coupling. Here, near the $D6$ end, the lowest monopole-sector masses scale as $M_r \propto \hbar\sqrt{|r|}/R_{S^2}$ (the lowest monopole harmonic at $j = |r|/2$ has eigenvalue $j(j+1) - r^2/4 = |r|/2$): a *square-root* tower. Linearity in $|r|$ holds near the $D7$ end, in the momentum regime; whether it survives deep collapse is precisely what supersymmetry guaranteed for Witten and nothing guarantees in this bosonic family. So: *collapse view* — a theorem; *emergence view* — Witten's reading minus the BPS protection, with a well-posed open problem in the gap: find, or exclude, a protection mechanism for the linear tower.

The gauge sector matches gratifyingly across the two readings. In $\text{IIA} \rightarrow \text{M}$ the Ramond–Ramond photon *enters from strong coupling* as R_{11} grows (the inertia rule again: $1/g^2 \propto R_{11}^3$); our $U(1)_{T_3}$ *exits to strong coupling* under collapse. Same phenomenon, opposite orientation — and Witten's case is the existence proof that the strong-coupling end of the coupling exit is physics, not pathology. Finally, the criticality itself is frame-dependent in a precise sense: seen from $D7$ the endpoint is a smooth infinite-distance limit; seen from $D6$ it is the nucleation of a new dimension

— one transition, read from the ordered or the disordered side.

7 What the 1980s knew, in six lines

1. **You cannot gauge what you quotient.** The circle you divide out has no Killing vector; its physics survives only as bundle data. (Where hypercharge went.)
2. **Couplings are inertias.** Ratios inside one simple factor are theorems; ratios across factors are moduli. (Why $1/4$ was ever predicted, and why the flip unpredicts it.)
3. **Charges are weights, and fermions see double covers.** One central element can lock parity ($q_8 \equiv 2I$), and through it, the SM's $\mathbb{Z}_6$. (Why the charge lattice came out right.)
4. **Scalars are locked; tensors and gravitini are not.** The Higgs can never be a scalar harmonic; the electron singlet lives at the canonical twist, in the gravitino. (Where the missing slots were hiding.)
5. **Odd dimensions are vector-like.** Chirality needs an even stage or a singularity; in this family it lives at $D6$ and must be transported. (The wall.)
6. **Dimensions die in two ways.** Vacuum to infinity, or coupling to infinity — and one geometric collapse can realize both at once, while charges convert smoothly to fluxes. (The modern question hiding in the old ladder.)

The verdict on the fork itself, for the record: Option B is structurally seductive (unbroken colour = unbroken base isometry; the charge lattice; the breaking pattern) and dynamically impossible on $X_{1,1}$ (no Y boson, no chirality); Option A is dynamically self-consistent and physically sterile (no colour at all). The geometry does not choose a labelling; it sells two different incomplete theories, and the price list — worked out line by line above — is the pedagogy. The constructive residue is a set of well-posed problems: the strong-coupling fate of the collapsing abelian, the hat-formation analysis on the moduli potential, the boundary/inflow formulation of the $D5 \leftrightarrow D6$ half, and the gravitino-borne electron as a feature to be tested in any future singular or fluxed completion.

A Representation data used in the text

Branchings under $SU(2) \times U(1)_8$ (subscript = q_8 , eigenvalue of $\text{diag}(1, 1, -2)$):

$$\mathbf{3} = \mathbf{2}_{+1} \oplus \mathbf{1}_{-2}, \quad \bar{\mathbf{3}} = \mathbf{2}_{-1} \oplus \mathbf{1}_{+2}, \quad \mathbf{8} = \mathbf{2}_{+3} \oplus (\mathbf{3} \oplus \mathbf{1})_0 \oplus \mathbf{2}_{-3}.$$

Tangent and spinor modules of $X_{1,1}$:

$$m = \mathbf{3}_0 \oplus \mathbf{2}_{+3} \oplus \mathbf{2}_{-3}, \quad \Delta_7 = \mathbf{2}_{+3} \oplus \mathbf{2}_{-3} \oplus \mathbf{3}_0 \oplus \mathbf{1}_0.$$

Gravitino module (the e_R carrier), by charge sector:

$$\Delta_7 \otimes m = (\mathbf{1} \oplus \mathbf{3})_{\pm 6} \oplus (\mathbf{3} \cdot \mathbf{2} \oplus \mathbf{2} \cdot \mathbf{4})_{\pm 3} \oplus (\mathbf{3} \cdot \mathbf{1} \oplus \mathbf{4} \cdot \mathbf{3} \oplus \mathbf{5})_0 \quad (\text{dim } 56),$$

with exactly one $SU(2)$ -singlet at each of $q_8 = \pm 6$. Three-form scalars:

$$\Lambda^3 m = (2 \cdot \mathbf{1} \oplus 2 \cdot \mathbf{3} \oplus \mathbf{5})_0 \oplus (2 \cdot \mathbf{2} \oplus \mathbf{4})_{\pm 3} \oplus \mathbf{3}_{\pm 6} \quad (\text{dim } 35),$$

with two doublet copies at each odd charge.

B The index in four lines

On $\mathbb{CP}^2$ with spin^c twist $c_1(L) = nH$ (n odd), $\hat{A} = 1 - p_1/24$, $p_1(\mathbb{CP}^2) = 3H^2$, $\int_{\mathbb{CP}^2} H^2 = 1$:

$$\text{ind } D_n = \int_{\mathbb{CP}^2} e^{nH/2} \left(1 - \frac{p_1}{24}\right) = \frac{n^2}{8} - \frac{1}{8} = \frac{n^2 - 1}{8} : \quad n = 3 \rightarrow 1, \quad n = 5 \rightarrow 3.$$

One generation at the canonical twist, three at the next odd flux: generation counting is literally counting flux quanta on the colour base. (Under Option B these zero modes are coloured — the index counts quark-like multiplets; the leptons ride the fibre spinor structure instead. The division of labour is itself a lesson.)

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Provenance. This document compiles the kkFable derivation notes N0–N12 (workspace **notes/**, ledger **notes/OPTION-B-VERDICTS.md**), which carry the full expansion blocks, counterargument passes, kill criteria, and two independent verification passes per load-bearing claim. Sources used beyond the inherited corpus are digested in **sources-local/**. No computer algebra was used in this run; the representation-theoretic identities were verified independently by a second model acting as a pencil-and-paper collaborator.